

GOSHEN, IN, USA

WMO: 724388

Lat: 41.525N

Lon: 85.791W

Elev: 253

StdP: 98.32

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.4	-22.0	0.5	-17.9	-19.1	0.7	-14.7	11.9	-3.4	11.1	-3.7	3.2	250	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.3	23.1	30.9	22.6	29.2	21.6	25.0	29.8	24.0	28.6	23.2	27.4	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.9	27.9	22.5	17.8	26.7	21.9	17.1	25.9	77.2	29.6	73.3	28.8	69.9	27.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.3	8.5	-22.4	34.2	3.5	1.9	-25.0	35.6	-27.0	36.7	-28.9	37.7	-31.5	39.1		
			-22.7	26.3	3.4	1.1	-25.2	27.1	-27.2	27.7	-29.1	28.3	-31.6	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-4.0	-2.8	3.2	9.9	15.9	20.9	22.5	21.5	18.1	11.4	5.0	-1.2
	DBStd	10.69	6.48	6.08	6.16	5.22	4.93	3.69	3.17	2.90	4.20	5.04	5.32	5.94
	HDD10.0	1652	435	358	227	65	7	0	0	0	1	43	166	349
	HDD18.3	3428	693	591	470	258	108	17	3	7	54	222	400	605
	CDD10.0	1692	1	2	16	61	190	328	388	356	244	87	16	2
	CDD18.3	425	0	0	1	4	33	94	133	104	48	8	0	0
	CDH23.3	3744	0	0	9	38	302	801	1233	859	441	61	0	0
	CDH26.7	1151	0	0	1	3	81	256	426	237	137	11	0	0
Wind	WSAvg	4.0	4.7	4.7	4.5	4.8	4.1	3.5	2.9	2.7	3.0	3.9	4.4	4.6
Precipitation	PrecAvg	909	48	43	65	88	90	98	94	95	87	70	71	63
	PrecMax	1346	146	136	152	176	211	241	250	297	253	190	152	134
	PrecMin	625	8	6	12	5	15	20	14	23	0	13	16	12
	PrecStd	145	30	28	31	37	42	50	50	53	52	44	33	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.3	24.5	27.1	31.7	33.5	35.0	32.7	32.8	28.2	21.2	16.1
		MCWB	12.2	11.8	17.2	17.4	21.3	22.4	24.8	23.9	22.3	20.6	14.5	13.9
	2%	DB	10.4	12.3	19.1	24.1	28.8	31.5	32.4	30.9	30.3	24.9	17.8	12.3
		MCWB	8.6	9.7	13.9	16.0	19.8	22.1	24.0	23.2	21.9	18.2	13.6	10.5
	5%	DB	7.3	8.5	16.0	22.0	26.9	29.6	30.9	29.1	27.9	22.4	15.6	9.1
		MCWB	5.3	6.4	11.7	14.8	19.1	21.2	22.9	22.2	20.4	16.8	12.2	7.3
	10%	DB	4.3	5.1	12.9	19.1	24.4	27.7	28.9	27.7	25.9	19.9	13.2	6.6
		MCWB	2.8	2.9	9.5	12.9	17.9	20.7	22.0	21.6	19.3	15.6	10.0	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.1	17.6	19.4	23.0	25.0	26.7	25.8	24.5	22.0	16.2	14.0
		MCDB	13.7	14.7	24.4	24.8	28.3	30.3	31.3	30.8	30.3	26.9	19.5	15.9
	2%	WB	9.0	9.9	14.9	17.3	21.4	23.8	25.4	24.6	23.1	19.7	14.5	10.5
		MCDB	10.0	12.1	18.7	21.8	26.7	28.9	30.2	28.8	27.9	22.4	17.0	12.0
	5%	WB	5.5	6.3	12.5	15.6	20.2	22.8	24.3	23.6	21.9	17.9	12.5	7.6
		MCDB	7.0	8.7	15.1	20.3	25.0	27.5	29.0	27.4	25.7	20.9	15.1	8.8
	10%	WB	2.9	3.2	9.6	14.0	18.9	21.8	23.3	22.6	20.7	16.1	10.4	5.2
		MCDB	4.3	5.0	12.9	18.8	23.2	26.0	27.6	26.2	24.2	19.5	13.2	6.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	9.7	11.7	11.2	10.8	11.3	11.2	12.0	10.5	8.6	6.9
		MCDBR	9.6	11.1	13.6	14.5	13.3	12.6	12.5	12.6	13.9	13.2	11.2	9.6
		MCWBR	8.4	9.2	9.1	8.3	7.0	5.5	5.6	6.0	6.4	7.4	8.2	8.4
	5% WB	MCDBR	9.1	10.9	12.2	12.7	11.5	10.2	10.7	10.3	11.3	11.1	10.1	8.9
		MCWBR	8.4	9.4	8.9	8.5	6.8	5.4	5.6	5.6	5.9	7.5	8.1	8.4
Clear-Sky Solar Irradiance	taub	0.299	0.332	0.351	0.398	0.434	0.436	0.440	0.444	0.408	0.353	0.318	0.298	
	taud	2.377	2.290	2.344	2.228	2.174	2.213	2.206	2.187	2.266	2.421	2.465	2.434	
	Ebn at Noon	850	868	896	872	845	842	834	818	823	838	823	822	
	Edh at Noon	82	105	112	135	146	141	141	139	120	91	75	72	
All-Sky Solar Radiation	RadAvg	1.50	2.33	3.50	4.62	5.29	6.00	6.08	5.30	4.35	2.85	1.78	1.26	
	RadStd	0.14	0.32	0.32	0.41	0.50	0.38	0.32	0.29	0.36	0.28	0.26	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page